

Multimedia

Vorlesung

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How Video Streaming Works

What are CDNs, HLS and DASH

In addition to the videos, which are linked on the next slides, this website contains a summary and details of the concepts discussed in the videos – find detailed information summarized in writing here:

<https://www.wowza.com/blog/streaming-protocols>

This helps with understanding when learning for the exam.

What Is A CDN? How Does It Work?

<https://www.youtube.com/watch?v=RI9np1LWzqw>

Relevant Exam Questions:

- What is a Content Delivery Network and what advantages does the usage of a CDN have?
- How does a CDN work and what is a PoP?
- What is DNS-based routing and what is Anycast?
- Why can a CDN provide content much faster?
- Why is a CDN more secure and more reliable?

How Does a Live Streaming Platform Work?

(YouTube live, Twitch, TikTok Live)

<https://www.youtube.com/watch?v=7AMRfNKwuYo>

Relevant Exam Questions:

- Give an example for a transport protocol for live streaming.
- What is Adaptive-Bitrate-Streaming?
- What is meant by segmentation in transcoding for video streaming?
- What are most common live streaming formats?
- What does an HLS file consist of?
- What means latency in video streaming and why is it relevant?

How does Netflix work? How does streaming work? Why is Netflix so fast?

<https://www.youtube.com/watch?v=G36Qbk0c2WM>

Relevant Exam Questions:

- How does Netflix increase speed even further by working with Internet Service Providers (ISPs)?

Live Streaming Protocol Comparison 1/2

<https://www.youtube.com/watch?v=jYYhwbF3tvM>

Details are also on the page

<https://www.wowza.com/blog/streaming-protocols>

Relevant Exam Questions 1/2:

- Is the same protocol and codec used for recording and playing a live streamed video?
- Which protocols are used on the recording/encoder side, which on the playback side?
- What is the main difference between TCP and UDP? (see also website <https://www.wowza.com/blog/streaming-protocols>)

Live Streaming Protocol Comparison 2/2

Relevant Exam Questions 2/2:

- What do you know about RTSP and what are the best applications to use it?
- Can you give details about Apple HLS?
- What is MPEG-DASH? Which codecs does it support? Which latency does it have?